

Representation Matters: Cross-Dataset Analysis of Implicit Hate Speech Detection Methods

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MOTIVATION

Implicit hate speech is one of the most challenging forms of online harm. Unlike explicit slurs, implicit hate relies on subtle phrasing, coded language, and context, making it difficult for both humans and automated systems to detect.

Key Challenges:

- 33% of online hate is implicit and is difficult to detect with traditional methods
- Poor cross-platform performance: models fail when applied to new communities
- Constantly evolving language where coded vocabulary changes rapidly

DATASET

We evaluate our approaches on three publicly available hate speech datasets with diverse characteristics and annotation schemes.

IHC [1]: Twitter hate communities dataset with 22K posts, including target and implied meaning annotations for implicit hate detection.

SBIC [2]: Social bias corpus with 45K posts featuring hierarchical offensiveness labels and stereotypical implication annotations.

DynaHate [3]: Dynamically generated dataset with 41K examples designed to test model robustness against deceptive hate speech.

Dataset	Train Set	Val Set	Test Set
IHC	12,234	4,078	4,079
SBIC	35,424	4,666	4,691
DynaHate	32,924	4,100	4,120
TOTAL	80,582	12,844	12,890

DATA PREPARATION

Dataset Processing:

- Label normalization - Convert all datasets to binary hate/not-hate classification
- Text preprocessing - Remove URLs, @mentions, normalize hashtags (#hate → hate)
- Data splits - 60/20/20 train/val/test, combined to 80/20 for final evaluation

Evaluation: Macro-F1 & AUPRC metrics for in-domain and cross-dataset validation

EDA (EXPLORATORY DATA ANALYSIS)

Metric	Value
Class balance	IHC: 60/40, SBIC: 70/30, DynaHate: 50/50
Dataset size	IHC: 20K, SBIC: 45K, DynaHate: 41K
Annotation	IHC: Human (bias), SBIC: Frames, DynaHate: Adversarial
Cross-domain	66.3% → 50.4% F1 (16% drop)

Key Patterns: Human annotation bias causes inconsistent labeling.

Challenges: Context dependency (SBIC social frames), cultural knowledge (IHC communities), adversarial deception (DynaHate).

Impact: Context-aware models handle cross-dataset challenges better.

IMPLICIT HATE EXAMPLE

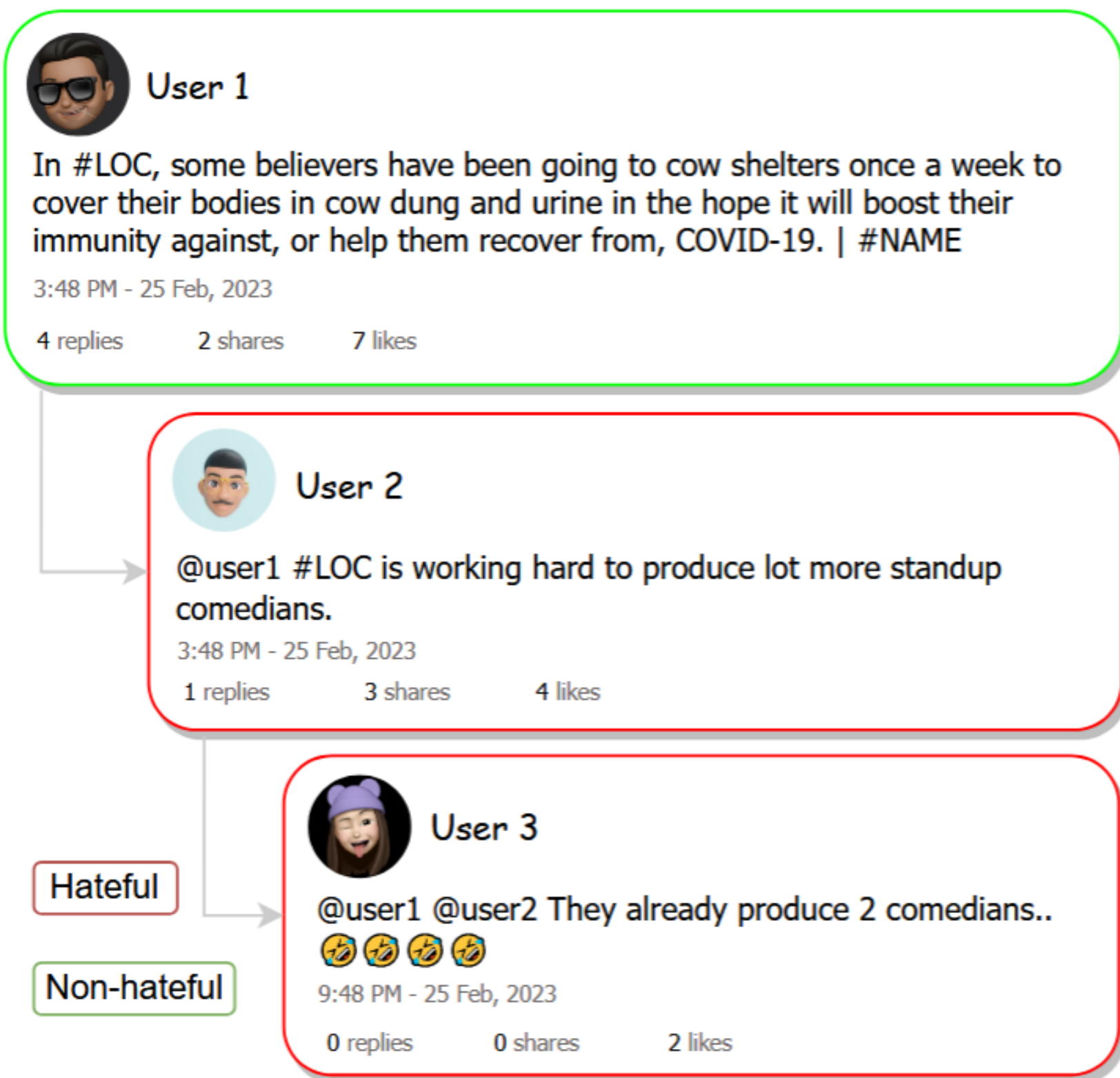


Figure: Social Media Thread with Implicit Hate [4]

APPROACHES

APP.1: Word EMB + SharedCon

- FastText/HateBERT [5] word embeddings for semantic representation
- Multi-pooling (max, avg, min) extracts diverse token-level features
- SharedCon contrastive learning + K-means clustering for classification

APP.2: Target-Aware Classifier

- Uses bag-of-words n-grams, POS bigrams, and 20-topic LDA
- Adds target-aware signals from NER entities, sentiment proximity, and ± 5 -token context
- Builds an 11,231-dimensional feature space and trains Linear SVM, Logistic Regression, and Multinomial NB

RESULTS & ANALYSIS

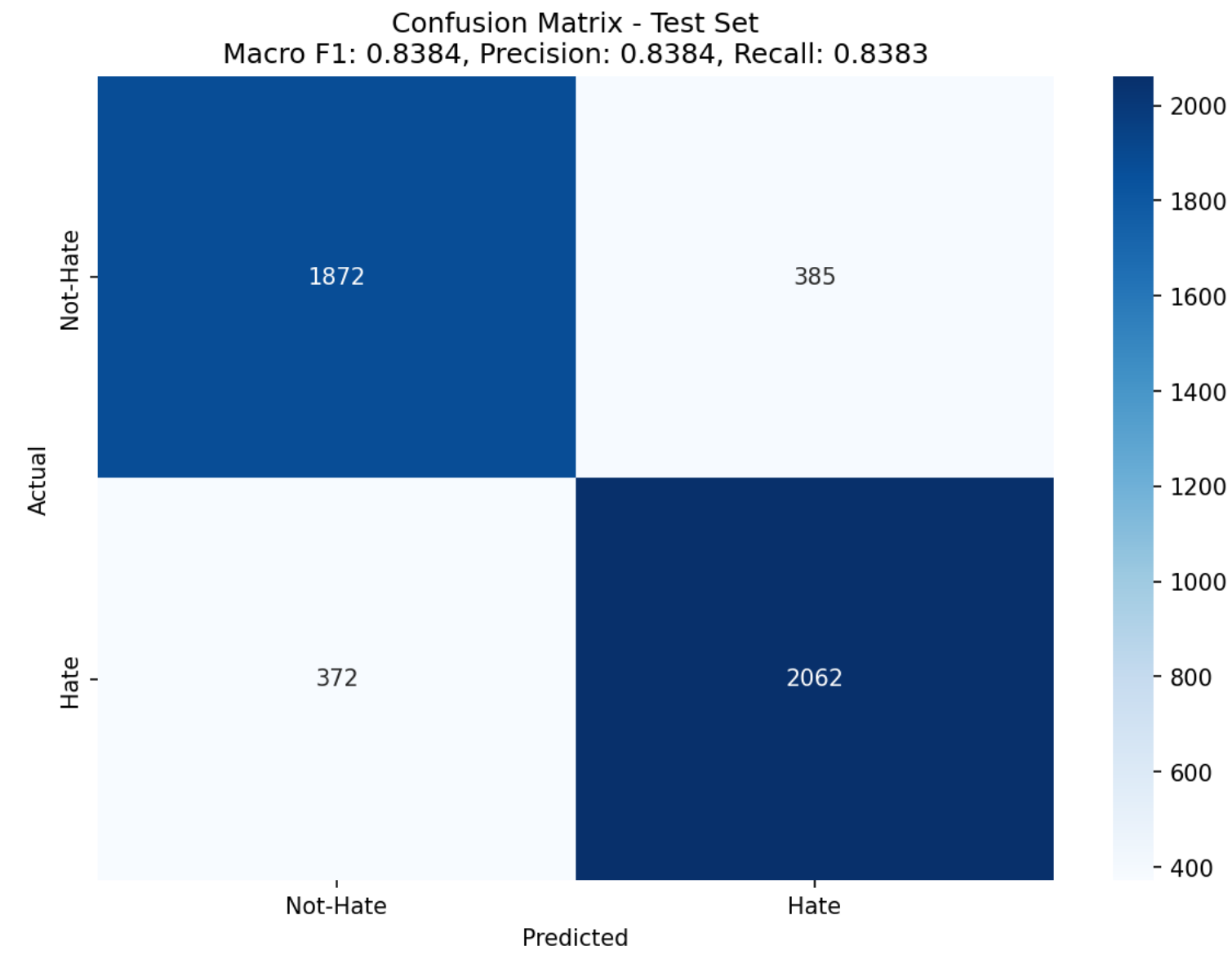
In-Dataset (F1 / AUPRC)

Dataset	App. 1	App. 2	App. 3	SharedCon
IHC	75.21/75.98	72.4/70.5	76.76/81.75	78.5/-
SBIC	83.84/92.25	79.8/89.0	84.18/93.15	84.3/-
DynaHate	79.72/90.58	61.7/73.6	81.35/91.55	79.1/-

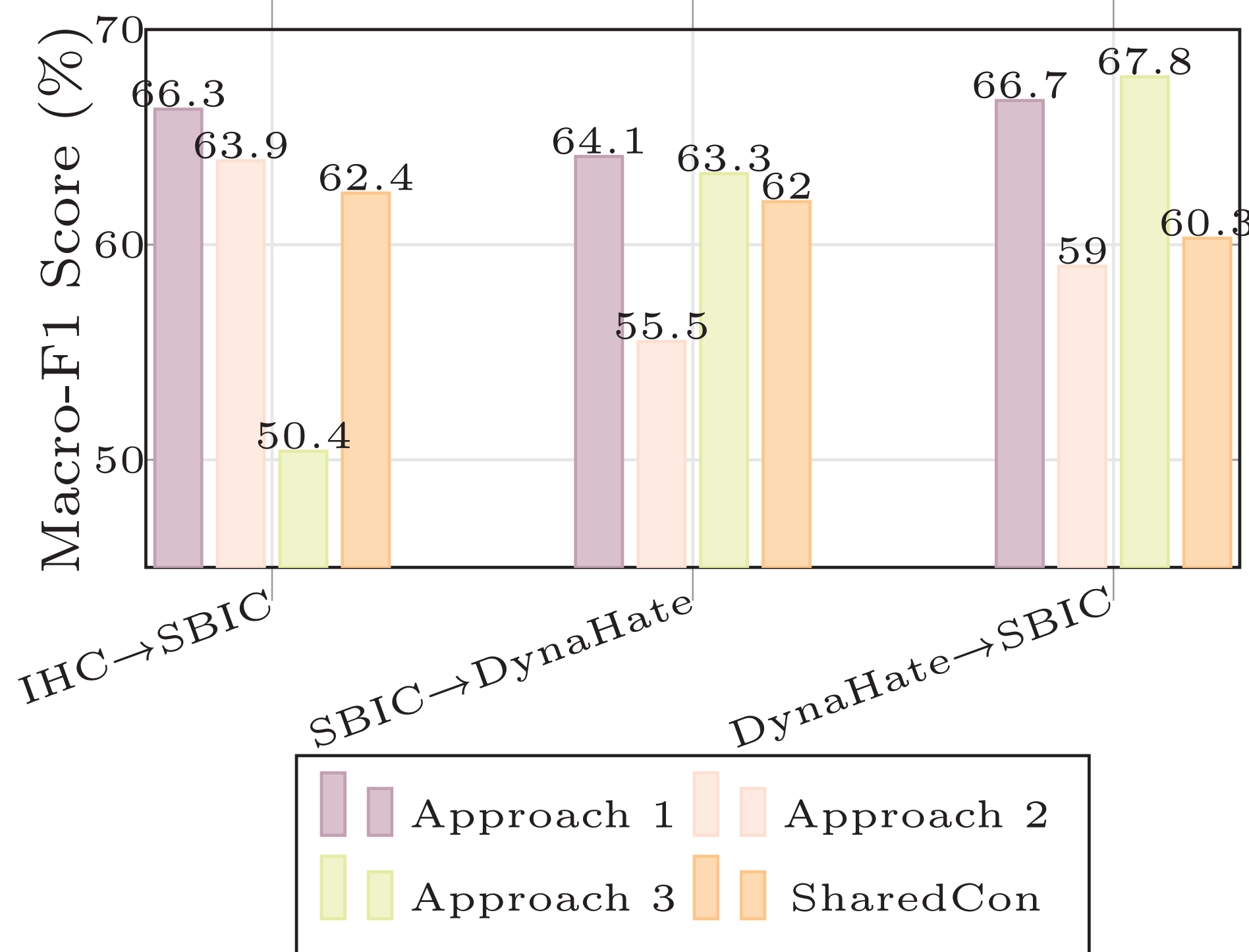
Cross-Dataset (F1)

Train → Test	App. 1	App. 2	App. 3	SharedCon
IHC → SBIC	66.31	63.9	50.43	62.4
IHC → DynaHate	53.32	49.8	44.70	67.0
SBIC → IHC	64.23	60.9	62.51	59.6
SBIC → DynaHate	64.13	55.5	63.27	62.0
DynaHate → IHC	64.12	59.6	65.31	59.5
DynaHate → SBIC	66.71	59.0	67.82	60.3

Best Performance Model



Generalization Chart



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